Practice workbook · Name heart chambers and major vessels from a diagram

Labelled heart or circulation diagrams asking for a chamber, a connected vessel, or properties of blood in it.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

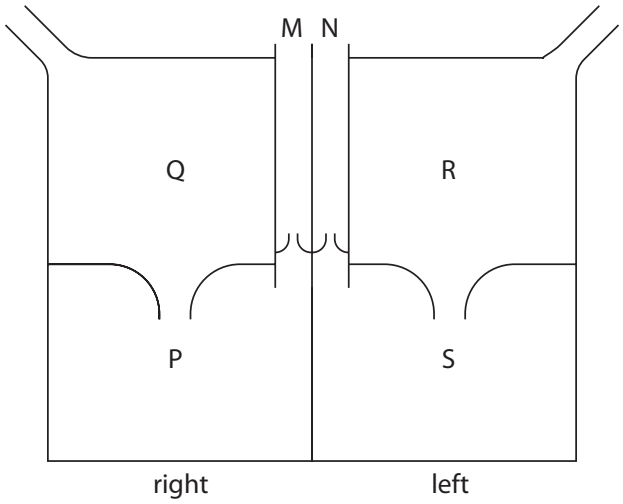
Type ID: HBI-NAME-HEART-CHAMBERS-AND-VESSELS

1. Question 1 · part a.i · 1 mark

January 2022 Paper 1

Worked-example retry · Attempt part a.i only.

1 The diagram shows a human heart in simple form.



(a) (i) Which of these is the name of blood vessel M?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) Which of these is the name of blood vessel N?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(iii) Draw arrows on the diagram to show the flow of blood through the heart.

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

1. Question 1 · part a.i · 1 mark · Working and answer

Working and answer

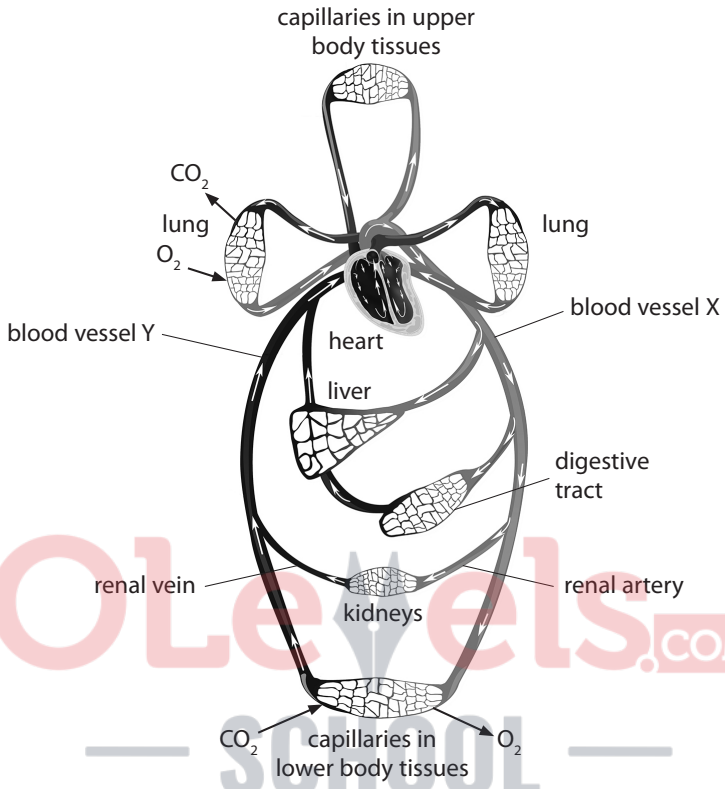
OLevels.com SCHOOL

2. Question 6 · part a.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.i only.

6 The diagram shows the human circulatory system.



(Source: © Designua/Shutterstock)

(a) (i) What is the name of blood vessel X?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 5 8 5 5 6 6 A 0 1 2 2 8

2. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

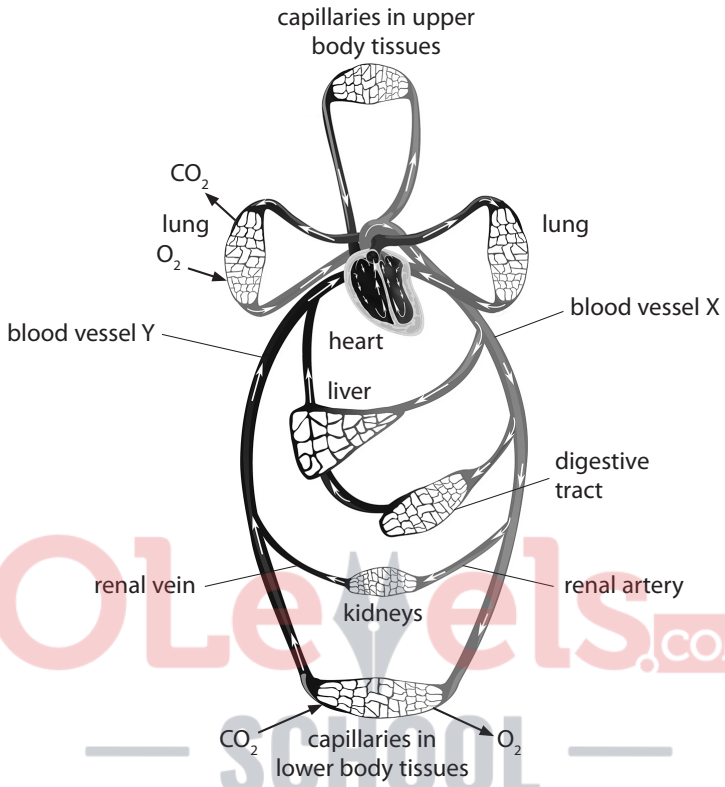
OLevels.com SCHOOL

3. Question 6 · part a.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part a.ii only.

6 The diagram shows the human circulatory system.



(Source: © Designua/Shutterstock)

(a) (i) What is the name of blood vessel X?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 5 8 5 5 6 6 A 0 1 2 2 8

3. Question 6 · part a.ii · 1 mark

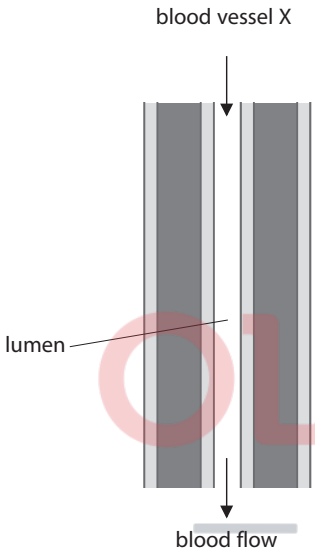
June 2019 Paper 2

Further practice · Attempt part a.ii only.

(iii) The diagram shows a longitudinal section through blood vessel X.

Draw a diagram to show a longitudinal section through blood vessel Y.

(3)



(iv) Explain the differences in the structures of blood vessel X and blood vessel Y.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

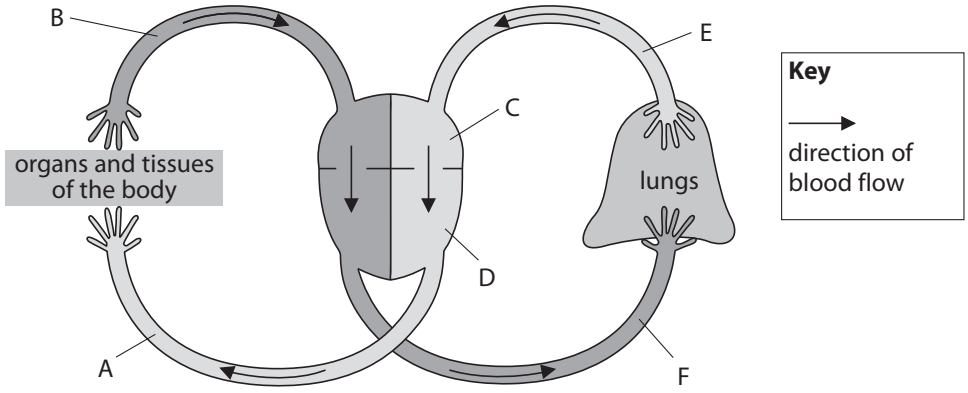
4. Question 4 · part a · 2 marks

January 2020 Paper 1

Further practice · Attempt part a only.

4 William Harvey was the first person to describe the circulation of blood around the body.

The diagram shows a model of the human circulatory system based on the research carried out by Harvey.



(a) Harvey suggested that each side of the heart was divided into two chambers.

State the names of chambers C and D.

(2)

chamber C .....

chamber D .....

(b) Harvey also suggested that blood is transported to and from the heart in different types of vessel.

Explain how the diagram supports Harvey's suggestion.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 5 9 9 0 9 A 0 1 0 2 8

4. Question 4 · part a · 2 marks · Working and answer

Working and answer

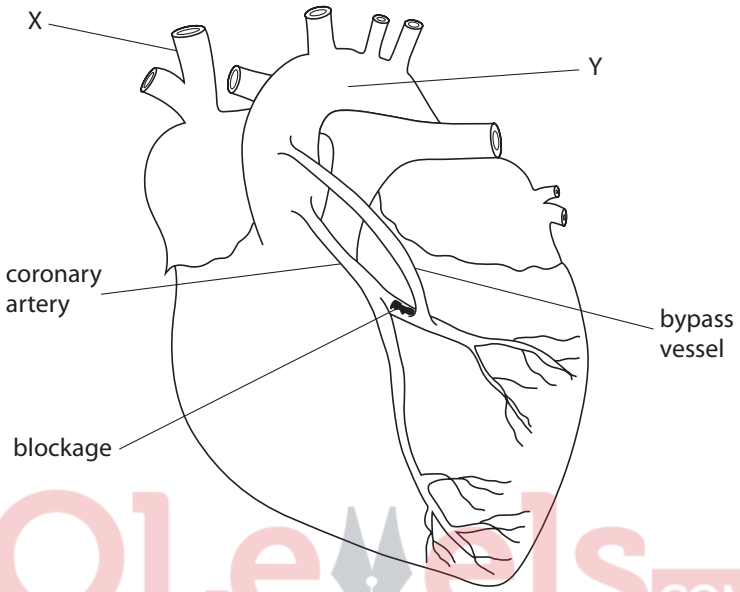
OLevels.com SCHOOL

5. Question 3 · part a.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part a.i only.

3 The diagram shows a person's heart. The person has had a heart bypass operation because a coronary artery had become partly blocked.



(a) (i) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) To which part of the heart is blood vessel X connected?

(1)

- ☒ **A** left atrium
- ☒ **B** left ventricle
- ☒ **C** right atrium
- ☒ **D** right ventricle

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

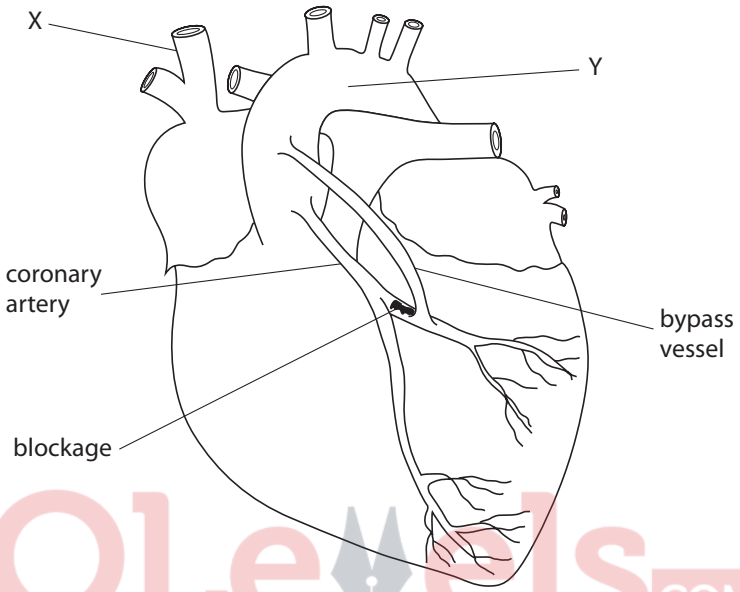
OLevels.com SCHOOL

6. Question 3 · part a.ii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

3 The diagram shows a person's heart. The person has had a heart bypass operation because a coronary artery had become partly blocked.



(a) (i) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) To which part of the heart is blood vessel X connected?

(1)

- ☒ **A** left atrium
- ☒ **B** left ventricle
- ☒ **C** right atrium
- ☒ **D** right ventricle

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6. Question 3 · part a.ii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

(b) (i) Before the bypass operation, the coronary artery was becoming blocked at the point shown on the diagram.

Draw a circle on the diagram to show the area of the heart that was affected by the blockage before the bypass operation.

(1)

(ii) Describe what effect this blockage would have on the heart tissue if the bypass vessel was not inserted.

(3)

(iii) Suggest how the bypass vessel allows the heart to work more effectively.

(2)

(Total for Question 3 = 8 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

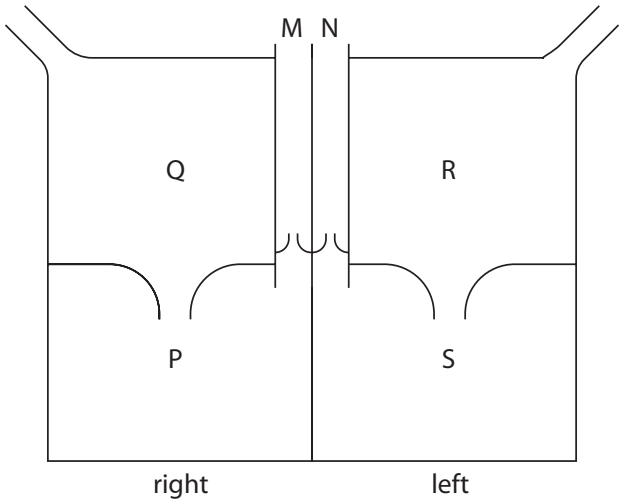
OLevels.com SCHOOL

7. Question 1 · part a.ii · 1 mark

January 2022 Paper 1

Further practice · Attempt part a.ii only.

1 The diagram shows a human heart in simple form.



(a) (i) Which of these is the name of blood vessel M?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) Which of these is the name of blood vessel N?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(iii) Draw arrows on the diagram to show the flow of blood through the heart.

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

7. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

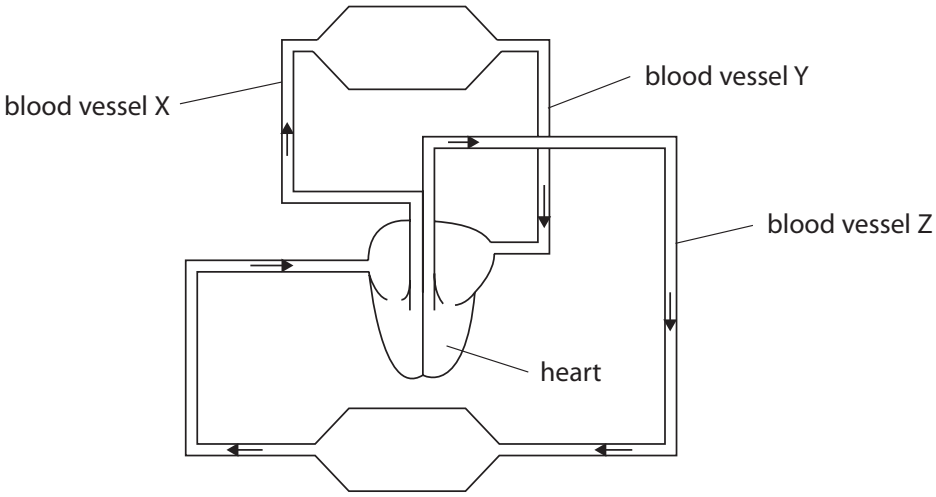
OLevels.com SCHOOL

8. Question 1 · part a · 1 mark

June 2022 Paper 2

Further practice · Attempt part a only.

1 The diagram shows a simple version of the human circulatory system.



(a) Name blood vessel X.

(1)

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 1 · part a · 1 mark

June 2022 Paper 2

Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) The box gives words about the circulatory system.

body	oxygenated	lungs	ventricle
	deoxygenated	atrium	

Complete the sentences using words from the box.

(5)

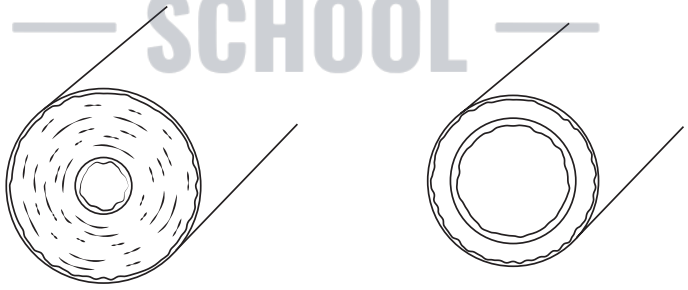
Blood vessel Y transports ..... blood from the

..... to the left ..... of the heart.

Blood vessel Z transports blood from the left ..... of the heart

to the .....

(c) The diagram shows the structure of blood vessel X and blood vessel Y.



blood vessel X blood vessel Y

Compare the structure of blood vessel X with the structure of blood vessel Y.

(3)

.....

.....

.....

.....

.....

.....

(Total for Question 1 = 9 marks)

8. Question 1 · part a · 1 mark · Working and answer

Working and answer

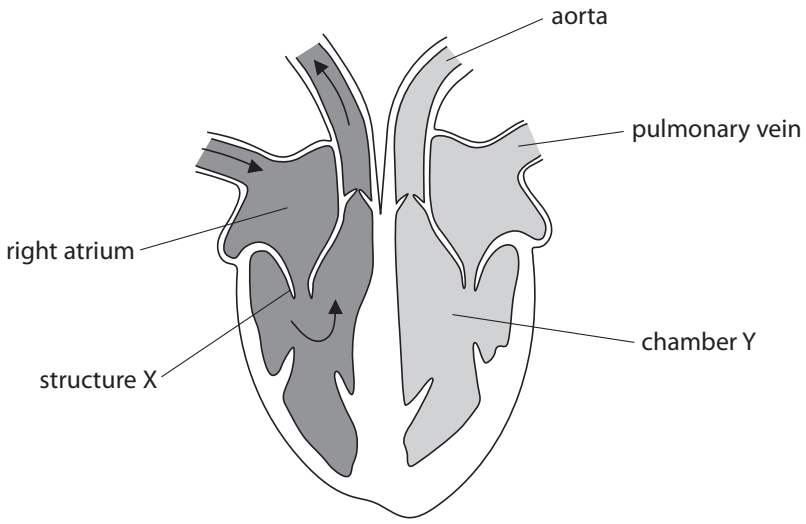
OLevels.com SCHOOL

9. Question 2 · part a.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part a.i only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 2 6 0 2 A 0 4 2 8

9. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

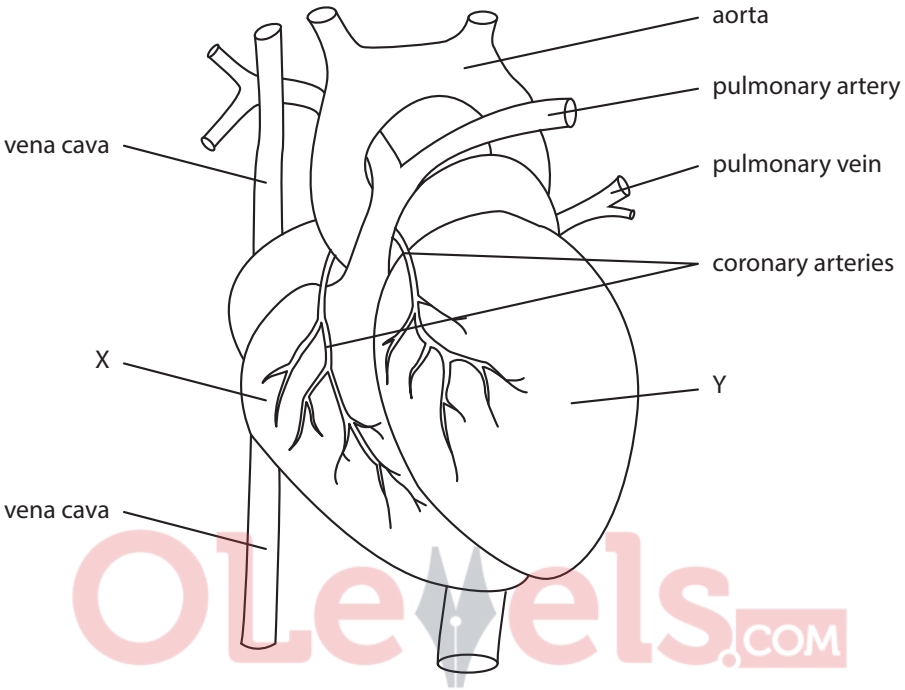
OLevels.com SCHOOL

10. Question 7 · part a · 1 mark

November 2023 Paper 1

Further practice · Attempt part a only.

7 The diagram shows the heart as seen from the front.



(a) The vena cava carries deoxygenated blood.

Name a different blood vessel, shown on the diagram, that also carries deoxygenated blood.

(1)

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10. Question 7 · part a · 1 mark

November 2023 Paper 1

Further practice · Attempt part a only.

(b) Blood chambers X and Y have walls that are different in thickness.

Explain this difference.

(5)

(c) The coronary arteries can become blocked.

Explain the effect of this blockage on the functioning of the heart.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 7 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

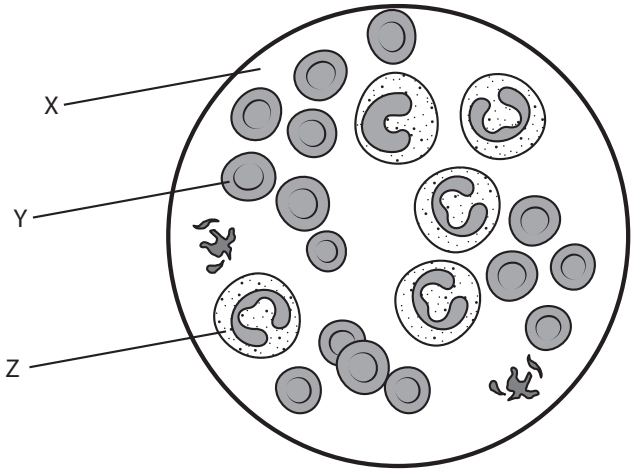
11. Question 2 · part c.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part c.i only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 (a) The diagram shows components, X, Y and Z, in a sample of human blood.



(i) Name components X, Y and Z.

(3)

X .....

Y .....

Z .....

OLevels.com SCHOOL

(ii) Describe the function of component Y.

(2)

.....

.....

.....

.....

(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

(1)

.....

.....

11. Question 2 · part c.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part c.i only.

(b) Blood is transported around the body inside blood vessels.

Draw **one** straight line from each blood vessel to a feature of the blood vessel.

(2)

Blood vessel	Feature of blood vessel
artery	contains valves to prevent backflow of blood
vein	has muscular walls to maintain high blood pressure
capillary	has thin walls for exchange of materials

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

11. Question 2 · part c.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part c.i only.

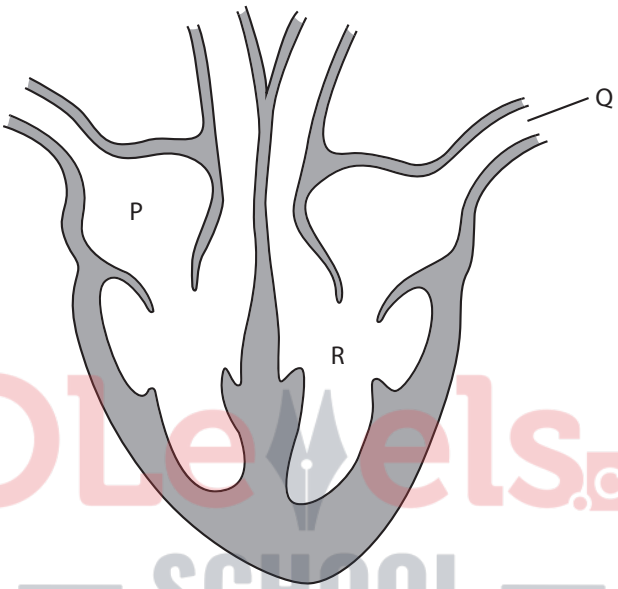
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) The heart pumps blood around the body.

The diagram shows a human heart.



OLevels.com SCHOOL

(i) The heart is divided into two halves.

Each half is divided into two chambers.

Name chamber R.

(1)

.....

(ii) Describe the route that the blood travels from part P to part Q.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question 2 = 13 marks)

11. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

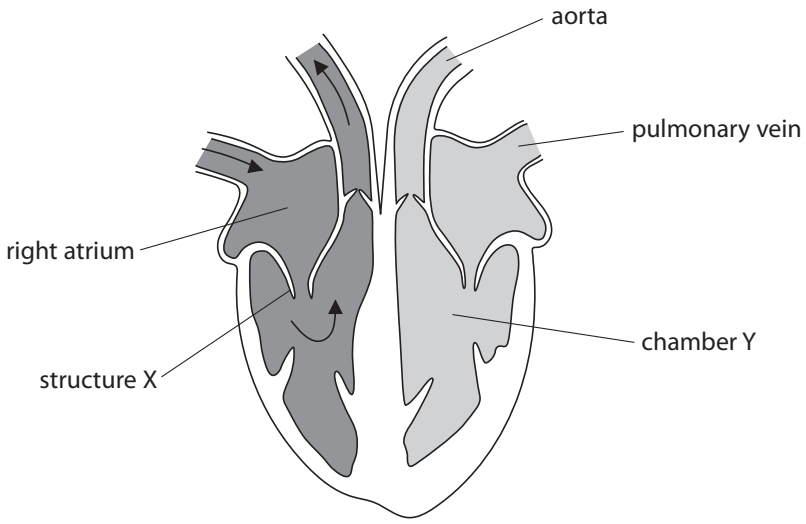
OLevels.com SCHOOL

12. Question 2 · part a.iv · 1 mark

January 2023 Paper 2

Further practice · Attempt part a.iv only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 2 6 0 2 A 0 4 2 8

12. Question 2 · part a.iv · 1 mark

January 2023 Paper 2

Further practice · Attempt part a.iv only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(iii) The table lists some properties of blood.

Add ticks (✓) to those boxes that show the properties of blood in the aorta.

(2)

Properties of blood	Blood in the aorta
oxygenated	
deoxygenated	
high pressure	
medium pressure	
low pressure	

(iv) State where blood in the left side of the heart has come from.

(1)

.....

(b) In one year in the United Kingdom, there were 173 heart transplants.

Heart transplant operations require hearts to be donated.

Some hearts cannot be used because the donor is too old.

Give two other reasons why a heart might not be suitable for a transplant.

(2)

1 .....

.....

2 .....

.....

OLevels.COM SCHOOL

12. Question 2 · part a.iv · 1 mark · Working and answer

Working and answer

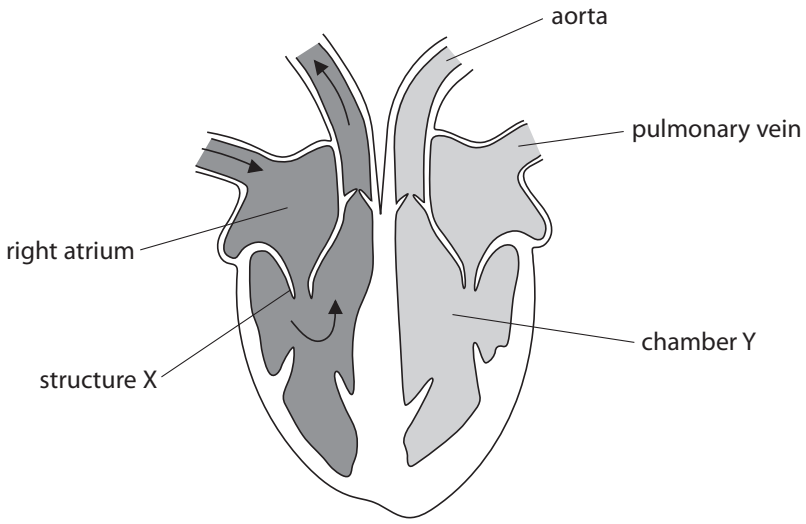
OLevels.com SCHOOL

13. Question 2 · part a.iii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.iii only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 2 6 0 2 A 0 4 2 8

13. Question 2 · part a.iii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(iii) The table lists some properties of blood.

Add ticks (✓) to those boxes that show the properties of blood in the aorta.

(2)

Properties of blood	Blood in the aorta
oxygenated	
deoxygenated	
high pressure	
medium pressure	
low pressure	

(iv) State where blood in the left side of the heart has come from.

(1)

.....

(b) In one year in the United Kingdom, there were 173 heart transplants.

Heart transplant operations require hearts to be donated.

Some hearts cannot be used because the donor is too old.

Give two other reasons why a heart might not be suitable for a transplant.

(2)

1 .....

.....

2 .....

.....

OLevels.COM SCHOOL

13. Question 2 · part a.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL